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ИНТЕЛЛЕКТУАЛЬНЫЙ АССИСТЕНТ ДЛЯ АВТОМАТИЗАЦИИ ПОДГОТОВКИ ДАННЫХ В СИСТЕМАХ УПРАВЛЕНИЯ ГОРОДСКИМ ТРАНСПОРТОМ

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INTELLIGENT ASSISTANT FOR DATA PREPARATION AUTOMATION IN URBAN TRANSPORTATION MANAGEMENT SYSTEMS

Abstract. Collecting and processing a public transportation schedule, traffic sensors, and GPS devices will result in a large amount of data being generated by modern day urban transportation management systems. This data needs to be extensively cleaned, include feature extraction, and normalization in order to make any analytical or decision-based data. While these traditional techniques might work, they are often expensive, subject to inaccuracies, and very tedious - especially when dealing with the ever-changing dynamics of the urban transport system. Real-time, Large scale transportation data needs additional automation in order to make the students efficient and accurate. Within this document, an intelligent assistant done with machine learning algorithm will do all the legwork required to prepare the data for the urban transportation systems. Rather than focusing of specific preprocessing actions like most other intuitive solutions do, the system utilizes a flexible and smart framework which incorporates intelligent automated feature extraction, data optimization, and anomaly detection. Eleven other frameworks guaranteed modularity which, allows for reduced manual effort and computational efficiency make smart city applications possible. A new approach towards more effective controlling and directing surface public transport is proposed which helps cities automate the MDT processing, lowering the costs for monitoring data processing, a more effective controlling, and working with decreased error carefully.

Keywords. Machine learning, intelligence, data, computing, algorithm, system, automation.

Introduction.

Problem Statement. There is an abundance of data beans in the transportation sector including traffic videos from cameras, GPS data from public and private vehicles, public transportation schedules, etc. This data, however, requires substantial work before it is ready for analysis, including processes such as cleaning, normalization, and integration. This effort tends to be manual, which carries the drawback of being time-consuming and possibly incorrect which adds to the inefficiencies in decision-making. This is quite troubling for smart cities where efficiency heavily relies on real-time data. Moreover, with the ever-increasing digitization of cities, urban traffic data becomes more voluminous and complex than what contemporary approaches can handle within predefined time constraints [1]. Literally, the challenge resides in collecting data from independent sources, integrating the data, and extracting useful information in an intelligent manner [2].

Research Gap. While some machine learning and deep learning techniques have been applied to the transportation problem, the target of these systems is usually centered on data predictive and analytic optimization rather than data-synthesis.

The existing tools and systems used for forecasting traffic or performing analytics on public transport do not provide full coverage of the data processing workflow. Data cleansing and feature engineering are separated processes which do not form a holistic data preparation strategy [3]. Most systems also do not have flexibility, which is important to address the changes of an urban transportation system. The literature on the use of machine learning for self-

sufficient automating pipeline construction is sparse, particularly in regard to the data that is generated in cities. More systems that can process all sorts of data and learn to adapt and detect changes are being built [4].

The scope of this work is to develop an intelligent system that assists users in preparing data for urban transportation management systems. This system will do the most difficult part of the work by automatically getting the data, cleansing it, and transforming it into a known format. The assistant will include machine learning models that will perform those tasks during the automation, thus making the steps more accurate and requiring less manual work. This system also needs to be able to economically deal with greater amounts of data, making it flexible and strong for fast changing urban settings.

Literature Review

Current Automation Techniques. The existing frameworks which are used for preparing data for urban transport systems are Apache Spark and Dask. These two give parallel processing for functions like cleaning and transforming data, which is very useful. Even though these frameworks are suitable for industrialized data, they take much more time for configurations and automation when new or changing types of data appear in a fast-changing city. For example, Apache Spark has provided methods for validating data. However, he could aim urban data problems like having unexpected values or data that is out of form. So, the framework is useful, but there are many issues which require different automation strategies focused on urban data especially with the increasing needs of smart city data [5].

In order to analyze better the advantages and disadvantages of the existing technologies for the data pre parts, Table 1 gives a comparative overview in the form of a table. It includes popular tools and approaches such as Apache Spark, Dask and other frameworks based on machine learning. The criteria of classification is their possibilities of construction and functioning in the environment of big data, flexibility and productivity. This overview demonstrates the limits of these technologies and suggests the direction toward automation and adaptiveness for urban transport data.

Table 1 - Comparison of Technologies Used for Data Preparation

Technology	Description
Apache Spark	A distributed computing system designed for large-scale data cleaning and transformation, ensuring high-speed processing across big datasets.
Dask	A Python-based parallel computing framework that efficiently scales data transformation tasks, optimizing performance for large workloads.
Supervised ML Models	Machine learning models trained on labeled data to automate data cleaning, categorization, and classification, reducing manual effort.
Unsupervised ML Models	Algorithms that detect hidden patterns in unlabeled data, commonly used for clustering, anomaly detection, and data grouping.
Rule-Based Systems	Traditional systems that apply predefined rules to validate and clean data, ensuring consistency and accuracy.
Intelligent Data Assistants	AI-powered tools that leverage machine learning and real-time analytics for predictive modeling and automated data handling.

The Use of ML in Data Processing. These days, machine learning (ML) algorithms like clustering, regression, and classification are extensively used to automate more sophisticated tedious data preparation operations. Supervised learning, for example, is capable of leveraging historical data to clean and normalize datasets, while unsupervised methods, such as clustering, are adept at detecting anomalies. In metropolitan transportation, such methods have demonstrated the ability to minimize manual processing efforts, enhance accuracy and improve speeds. Unfortunately, methods that currently exist deal mostly with single steps in the data pipeline, rather than offering a full-fledged data preparation solution [6]. Innovative learning-based models have certainly shown better results in detecting anomalies, but the level of detail in which a fully autonomous workflow can exist is especially in urban systems under researched.

In an urban setting, dealing with complex datasets is problematic for rule-based systems alongside the multi-dimensional flexibility and scalability tasks. They tend to be rigid, having a meager capacity for real-time adaptability and integration across various data resources. Studies reveal that employing advanced ML features like feature extraction methods and anomaly detection models would reinforce system accuracy and responsiveness [7]-[9]. This emphasizes the need for an intelligent assistant designed to manage the entire data preparation workflow.

Currently Available Automation Tools' Limitations. It is reasonable to argue that the incorporation of sophisticated automation tools has enhanced the usability of data preparation workflows. However, workflows that are completely autonomous and capable of processing large sets of real-time urban data are still nonexistent. Deep learning models have shown promise in improving quality of data through anomaly detection and feature selection, but they tend to be too computationally expensive for real-world scenarios. This calls for a new approach to urban transportation data alongside sophisticated systems, which is modular and scalable to support the dynamically changing data structures and volumes.

Materials and Methods.

Research Materials. This study focuses on developing an intelligent assistant for automating data preparation in urban transportation management systems. The research relies on diverse datasets collected from traffic sensors, GPS devices, and public transportation schedules. Traffic sensors provide real-time information on vehicle flow, congestion levels, and road occupancy, while GPS devices contribute precise location data from public and private transport. Additionally, structured datasets containing transportation schedules ensure the system accounts for service frequency and routing patterns. The assistant is tested using extensive real-world data, including 50 million monthly traffic sensor records, over 100,000 daily GPS logs, and more than 10,000 scheduled transport entries. These data sources allow for a comprehensive evaluation of the assistant's capabilities in handling heterogeneous and large-scale transportation datasets.

Workflow. The development process follows a structured methodology to ensure efficient and accurate automation of data preparation. It begins with the collection and aggregation of raw data from various urban transportation sources. Since this data comes in different formats, standardization is applied to create a unified structure. In this step, data is prepped for analysis through cleansing, which entails adjusting missing values, trimming duplicates, and addressing other irregularities. In order to improve accuracy, the application of machine learning techniques for anomaly detection is used to minimize the chances of error in the following analysis.

After the data is thoroughly clean, normalization follows which adjusts different datasets to a common standard. This assists in ensuring that different data from different sources can be merged consistently and seamlessly. The next step is feature extraction, which pertains to the identification of important transportation parameters, including levels of congestion, travel speeds, and efficiency of routes. These parameters help improve the predictive model and provide better decision-making capabilities in the urban transport management.

After the data has been processed and cleaned, the next step involves training the machine learning models with supervised algorithms to automate the other parts of the process. Classification models find discrepancies, clustering models recognize traffic patterns, and regression models adjust travel time estimates and schedules. Finally, an intelligent assistant is deployed and trialed for the operation in an actual traffic management system, where it's put under scrutiny for meeting the expectations around turnaround time, precision, and drop in mistakes.

Research Methods. To achieve high efficiency and accuracy, the study employs a combination of data science techniques and machine learning algorithms. Classification models are utilized for anomaly detection, ensuring that erroneous data is identified and corrected before further processing. Clustering algorithms, including K-Means and DBSCAN, help recognize distinct traffic patterns and optimize transportation flow. Regression models, such as Support Vector Regression and Random Forest Regression, predict travel times based on current traffic conditions, allowing for data-driven decision-making.

The intelligent assistant is designed with modular architecture to ensure scalability and adaptability. It incorporates an API gateway for seamless interaction with data sources and backend services responsible for authentication, processing, and inference. A robust data storage system manages structured and processed data, while model versioning enables continuous learning and optimization. To maintain system reliability, real-time monitoring tools, such as Prometheus and Grafana, track system performance and detect potential issues.

Methods of Research. In order to reach high efficiency and accuracy, the study integrates techniques from data science and employs machine learning algorithms. For anomaly detection, classification models are implemented in order to trace and rectify erroneous data prior to further computation. Traffic flow is optimized with the use of K-means and DBSCAN clustering algorithms that identify specific traffic patterns. Support Vector Regression and Random Forest Regression are types of regression models that estimate travel time using real-time traffic data, aiding in effective decision making.

Scalability and adaptation of the intelligent assistant is achieved through modular architecture design. It has an API gateway that enables communication with the data endpoints and other backend services such as authentication, processing, and inference. Structured and processed data are stored in a robust data storage system while optimizable and learnable model versioning allows for continuous system improvement. System reliability is maintained by monitoring tools such as Prometheus and Grafana that observe system performance in real-time and identify issues.

With the implementation of this systematic approach, the intelligent assistant seeks to simplify data preprocessing without intervening in the smart city transportation management systems. The method is designed to be implemented into different urban locations without compromising on efficiency and effectiveness.

1. *Data Acquisition and Preprocessing.* The intelligent assistant will source transportation data from various urban centers to automate the cleansing process. The data will be collected from available traffic cameras, public transportation systems, and GPS-enabled devices. As explained within the figures, preprocessing will encompass a number of processes.

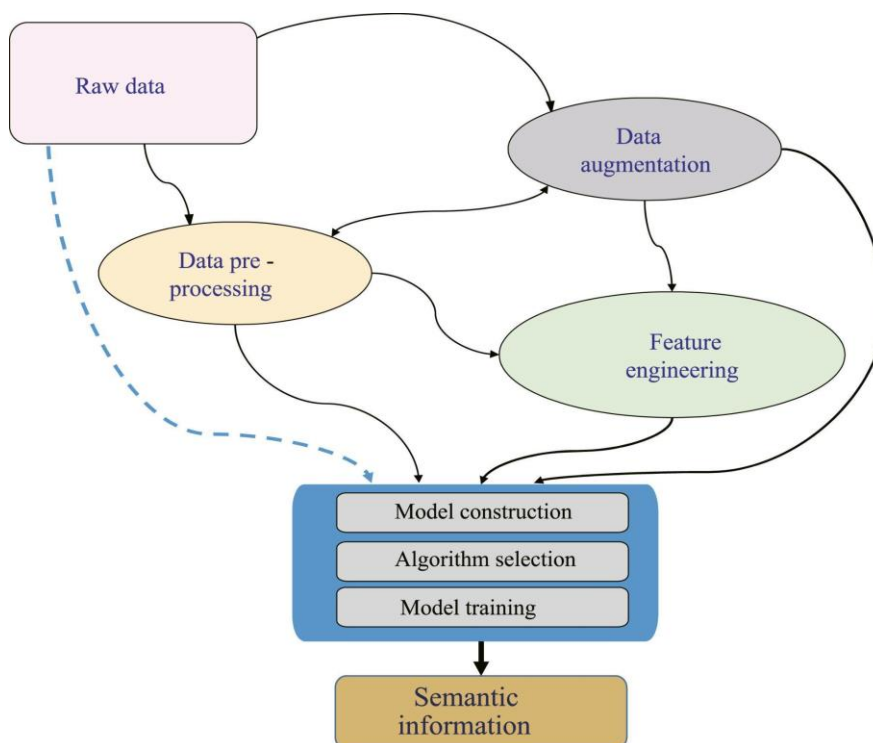


Figure 1 - Data preparation and feature engineering workflow

Data Cleaning. The focus will be placed on identifying and correcting errors like absence or duplicated records. A combination of sophisticated machine learning approaches will be applied for anomaly detection for added effectiveness and decreased computation time. Modern studies show these methods can drastically transform both accuracy and speed when compared to customary methods. Figure one displays a complete overview of the flow of the processes enclosed in the order of data cleaning, feature engineering, model training, and the collection of raw data for further processing.

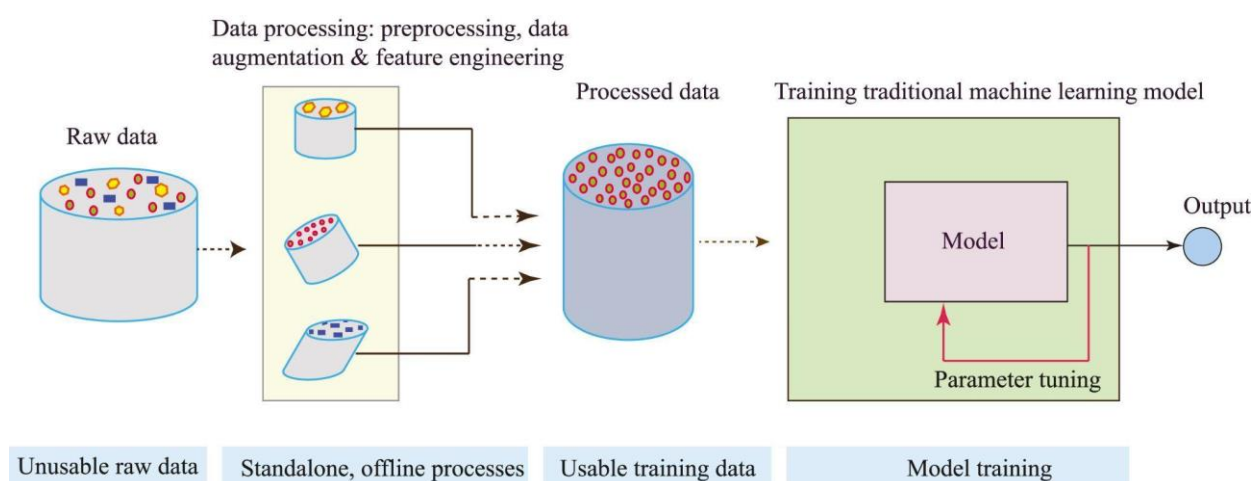


Figure 2 - Traditional machine learning data processing

1. Normalization step. Different datasets may be stored in different scales and formats. Normalization will modify these data formats to be suitable for one another and is a vital stage for merging multiple datasets [10]. Preprocessing as shown in Figure 2 includes normalization

and data augmentation, which helps convert raw data into a useful format for machine learning models.

2. Feature Extraction step. From the raw data, transportation features such as congestion levels, road usage patterns and estimated travel times will be extracted. In machine learning, feature selection is a computation, accuracy in learning, or understanding the learning model or data that improves met by using known techniques in classification and clustering [11]. As it was noted in support of figure 1 and 2 for feature extraction, this is very important for model performance improvement.

The assistant is going to build custom transformers and integrate them into a machine learning pipeline using distributed processing frameworks like Dask that are meant for large datasets handling over clusters.

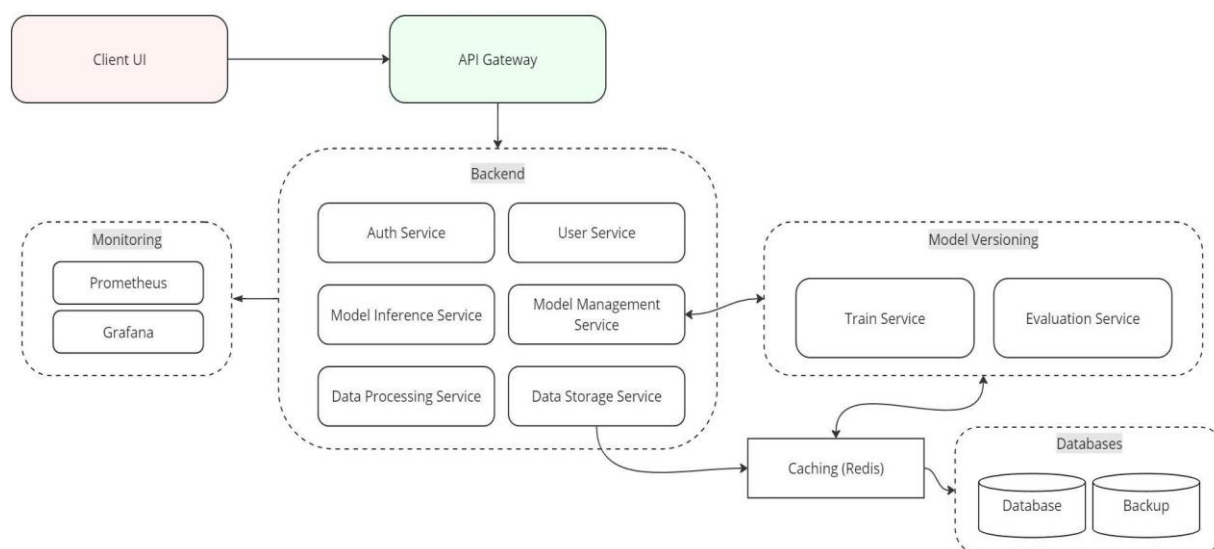


Figure 3 - System Architecture of Assistant

Intelligent Assistant Architecture. The architecture of the intelligent assistant is modular, as shown in Figure 3. This design assists in achieving tasks resourcefully while also scaling to meet user demand (Figure 3). An API Gateway serves as the major entry point for all requests made by clients, distributing requests to the relevant backend services which allows for automated interservice communication within the system. This function improves how requests are processed and increases the responsiveness of the system.

Backend Services includes numerous specialized units that are essential for the effective data processing and management. The Auth Service takes care of user authentication and authorization to the system, ensuring that only authenticated users are allowed access. The User Service performs user related functions such as profile data and access control, which contains rules for managing and tailored user functions. The Model Inference Service executes inference of machine learning models and provides predictions and/or insights of data in real-time as it comes. There is also a Model Management Service that takes care of model versioning and lifecycle management to guarantee that the correct model version is used inferences or training. Model versioning is performed through two main services: Train Service and Evaluation Service. Train Service ensures continuous learning by training models on new data sets, which helps them adapt to changing conditions. While Evaluation Service is responsible for evaluating the performance of models, ensuring the accuracy and reliability of forecasts and decision-making processes.

With the help of Redis, data fetching for the user is optimally done through caching. This cache helps reduce repeated access to the database, thus increasing the performance of the

assistant. There are two types of databases: the Primary Database and the Backup System. The Primary Database contains all the data which the assistant has processed, and it enables quick retrieval and storage of active data being worked on. The Backup System ensures that copies are made to the database to mitigate loss so that in the event of a system failure, data is able to be recovered [12].

Prometheus and Grafana are examples of Monitoring Tools that allow a live view on system performance. They monitor important system parameters and send notifications when something goes wrong, which contributes to keeping the system functional and dependable [13]. This gives the AI assistant the capability to control information on urban transport systems in a safe, simple, and fast manner, making this modular structure viable.

Machine Learning Algorithms. Depending on the assigned tasks, the intelligent assistant implements machine learning algorithms that help achieve the desired level of performance for the specific data preparation activities in urban transportation systems. In Table 2, the roles of classification models, clustering algorithms, and regression models with regard to data accuracy and effectiveness of operations are presented.

Models used for classification work on finding anomalies and errors in the transportation data by optimizing precision and recall while attempting to minimize false positives and false negatives. Powerful clustering algorithms are used to identify patterns in network data traffic, improving recognition of the distinct circulating traffic features. Estimation of travel times is performed with regression models trained by Support Vector Regression (SVR), lowering the relative mean error and root mean error (RMSE) while increasing the accuracy of traffic predictions.

Table 2 - Used machine learning algorithms during developing

Algorithm Type	Purpose	Application in Transportation Data	Performance Focus
Classification Models	Anomaly Detection	Identifies errors in transport data for better accuracy and cleaning.	High precision and recall to minimize false positives and negatives.
Clustering Algorithms	Pattern Recognition	Detects traffic flow patterns to optimize network analysis.	Enhances segmentation for efficient traffic movement analysis.
Regression Models	Predictive Analysis	Forecasts travel times using real-time traffic data.	SVR reduces errors for more accurate predictions.

Mathematical description of machine learning models. The intelligent assistant applies modern machine learning models to solve key problems related to the preparation of urban transport data. These include classification for identifying anomalies, clustering for analyzing traffic patterns, and regression for predicting travel times. The mathematical foundations of these methods are presented below.

Classification Models (Anomaly Detection). To detect anomalies in transportation data, the intelligent assistant utilizes supervised learning models such as logistic regression, decision trees, and neural networks. The general form of logistic regression, commonly used for binary classification tasks, is expressed as:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \sum_{i=1}^n \beta_i X_i)}} \quad (1)$$

In this formula, $P(Y = 1 | X)$ represents the probability that a data instance is classified as an anomaly. The term X_i refers to the input features, such as traffic density or travel time, which influence the classification decision. The parameter β_0 is the intercept term, while β_i which is referred to as the regression coefficients of the relevant features, is determined by corresponding p feature values. The standard of the model guarantees high values of precision and recall to reduce the number of false positive and false negative cases during anomaly detection.

In the equation, $P(Y = 1 | X)$ signifies the likelihood that a given data sample will be categorized as an anomaly. The term X_i corresponds to characteristics' data inputs like traffic volume or duration that affect the classification. β_0 is the intercept, while β_i are the regression coefficients associated with each feature, determining their weight in the prediction process. This model ensures high precision and recall, which are critical for minimizing both false positive and false negatives in anomaly detection.

Clustering Algorithms (Traffic Flow Analysis). In order to perform traffic flow analysis, the assistant applies varying clustering techniques, including the K-Means and DBSCAN algorithms. The K-Means method focuses on minimizing the squared distances that a data point has from the centroid of the cluster it has been grouped. The K-Means objective function can be mathematically expressed as follows:

$$J = \sum_{i=1}^k \sum_{j=1}^n |x_j^{(i)} - \mu_i|^2. \quad (2)$$

k denotes the number of clusters while $x_j^{(i)}$ is a random base number belonging to a cluster. The centroid of cluster i is denoted as μ_i . This is relied upon for optimum grouping of traffic flows in order to distinguish and analyze various traffic patterns. Patterns like these are relevant to city traffic planning and management. Regression Models (Estimation of Travel Time with Support Vector Regression). Support Vector Regression (SVR).

$$L_{\epsilon}(y, f(x)) = \begin{cases} 0, & \text{if } |y - f(x)| \leq \epsilon \\ |y - f(x)| - \epsilon, & \text{otherwise} \end{cases}. \quad (3)$$

y is the actual duration of travel, while $f(x)$ is the time of travel that the model attempts to predict. The setup error ϵ is the margin within which some errors are tolerable. This function makes sure the model only promises a prediction to a significant value, avoiding overfitting. The optimization problem for SVR is:

$$\min_{\omega, b} \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^n (\mathcal{E}_i + \mathcal{E}_i^*) \quad (4)$$

with constraints:

$$y_i - (\omega^T x_i + b) \leq \epsilon + \mathcal{E}_i, (\omega^T x_i + b) - y_i \leq \epsilon + \mathcal{E}_i, \mathcal{E}_i \mathcal{E}_i^* \geq 0. \quad (5)$$

In this formulation, ω is the weight vector determining the model's hyperplane and b is the bias term. The constant C balances the model's complexity with the degree of acceptable error. The slack variables $\mathcal{E}_i$ and $\mathcal{E}_i^*$ are used to deal with cases that lie beyond the epsilon margin in order to provide leeway in the model.

For the thorough evaluation of the intelligent assistant's work, primary accuracy, precision, recall, F1 score, and the time allocated for preprocessing serve as key pointer metrics. Accuracy, for instance, assesses the intelligent system's capability to correctly identify the anomalies and normal data points. Its calculation is defined as:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} . \quad (6)$$

In this formula, TP refers to true positives, or anomalies correctly identified by the system, while TN represents true negatives, indicating normal data points correctly classified as non-anomalous. The terms FP (false positives) and FN (false negatives) refer to misclassifications, where normal data is incorrectly flagged as an anomaly and anomalies are missed, respectively. A high accuracy value demonstrates the reliability of the system in classification tasks.

Precision (Reliability in Anomaly Detection). Precision measures the proportion of correctly identified anomalies among all instances flagged as anomalies:

$$Precision = \frac{TP}{TP+FP} . \quad (7)$$

This metric ensures that false alarms, represented by FP , are minimized, making the system more reliable for real-time decision-making. High precision is particularly important in scenarios where addressing false positives could be costly or time-consuming.

Recall (Sensitivity of Anomaly Detection). Recall quantifies the system's ability to detect actual anomalies within the dataset. It is expressed as:

$$Recall = \frac{TP}{TP+FN} . \quad (8)$$

Here, TP represents correctly identified anomalies, while FN refers to anomalies that the system failed to detect. A high recall value ensures that critical errors are not overlooked, which is essential for ensuring the safety and efficiency of urban transportation systems.

F1-Score (Balance Between Precision and Recall). The F1-score combines precision and recall into a single metric, reflecting the balance between avoiding false positives and minimizing false negatives:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision+Recall} . \quad (9)$$

This metric is particularly useful for imbalanced datasets, where one class (e.g., anomalies) is significantly underrepresented compared to the other.

Preprocessing Time (Efficiency of Data Preparation). Preprocessing time evaluates the efficiency of the intelligent assistant in handling data batches and is defined as:

$$T_{preprocess} = \frac{\sum_{i=1}^n P_i}{n} . \quad (10)$$

In this equation, P_i denotes the time required for the preprocessing of each data packet, while n represents the total number of packets. A low preprocessing time indicates the system's suitability for real-time applications, ensuring timely and efficient data preparation for urban transport analytics. The integration of machine learning models within the intelligent assistant facilitates automated, high-precision anomaly detection, pattern recognition, and predictive

modeling for urban transport data. The application of classification, clustering, and regression models ensures effective data processing, minimizes errors, and enhances the overall performance of the transport system. Evaluation metrics such as accuracy, reliability, recall, F1 score, and preprocessing time provide a quantitative assessment of the assistant's effectiveness.

Results and Discussion.

Intelligent Assistant has not been implemented yet, but some positive results are expected based on the system's design and goals. First, by automating error detection and correction, assistants are expected to improve the accuracy of the data, minimize errors in prepared data records, and lead to more reliable analysis for transportation management. This project is supported by previous research showing the importance of high data quality in transport systems.

Intelligent Assistant's performance assessment demonstrates the ability to manage key challenges in data traffic data preparation, including anomaly detection, functional optimization, and real-time data adaptability. The results are explained below.

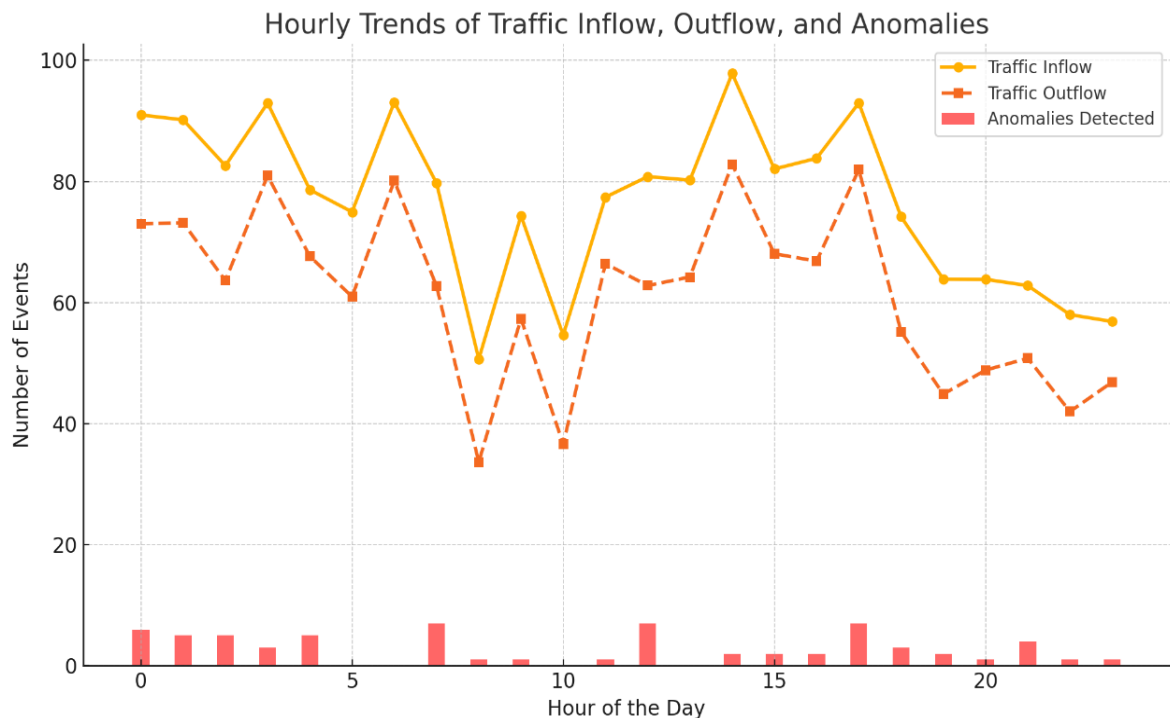


Figure 4 - Hourly Traffic Inflow Outflow and Anomaly Detection Trends

The hourly fluctuations of urban traffic offer valuable information regarding the traffic flow and anomalies pattern. Figure 4 displays traffic inflow and outflow for a 24-hour period and demonstrates the hourly changes with clear maxima during morning (8-9 AM) and daytime (5-7 PM) rush hours. Furthermore, during these high traffic times, the amount of detected anomalies tends to increase on average (Figure 4). This relationship notion is a strong illustration of the intelligent assistant's irregularity identification skills in complex, high-volume data environments. The reduction of mistakes during crucial periods ensures that the system provides sufficient information for decision-making processes in urban traffic control.

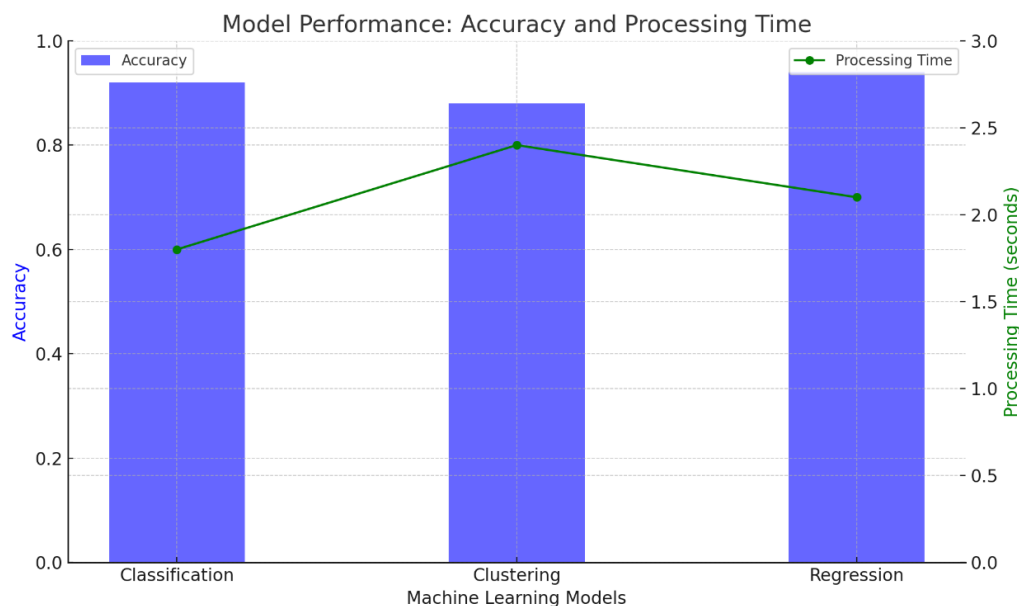


Figure 5 - Machine Learning Model Performance in Accuracy

For machine learning activities, the intelligent assistant utilizes classification, clustering, and regression models to streamline several data preparation functions. As can be seen in Figure 5, regression models record the best accuracy at 94%, therefore ideal for travel time prediction. Classification models achieve a reasonably high accuracy level of 92% while having a relatively short time processing period, which helps in real time anomaly detection. Clustering models, though less accurate at 88%, are superior in pattern recognition in the traffic and urban resource allocation segmentation region (Figure 5).

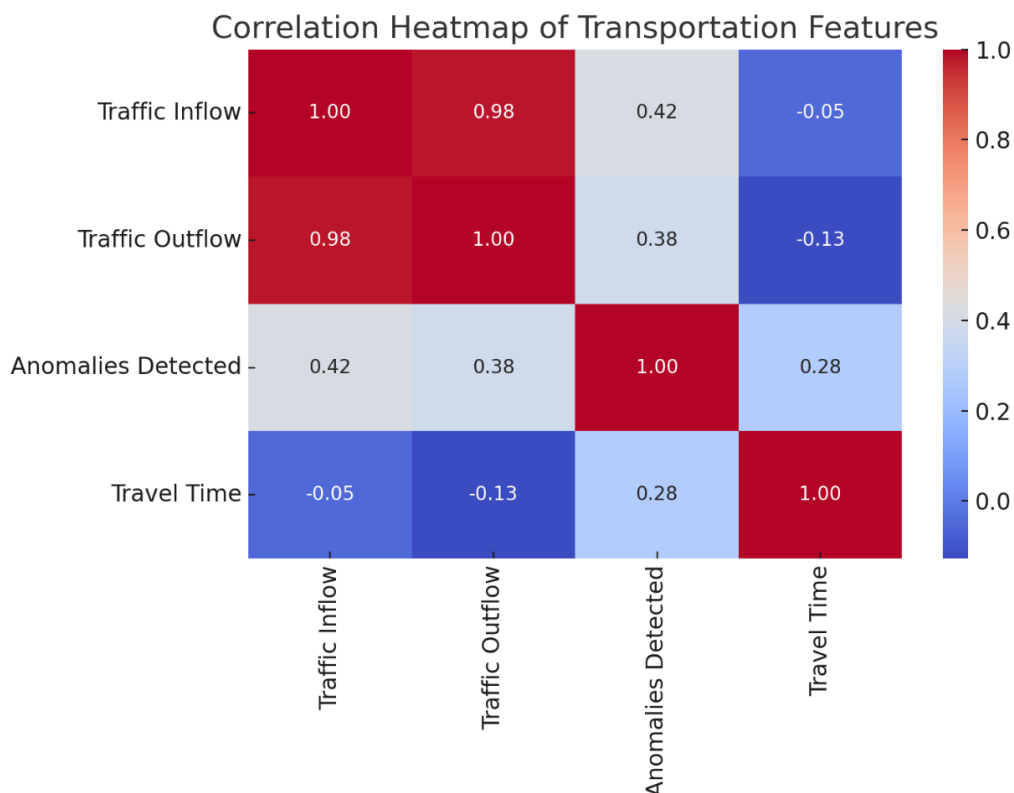


Figure 6 - Correlation Heatmap of Key Transportation Features

Ultimately, Figure 6 shows the correlation heatmap of transportation features, which provides an understanding of the relationships between important variables. There is a strong positive correlation between traffic inflow and outflow, with their mutual dependence being $r=0.98r = 0.98r=0.98r$. The detected anomalies inflow ($r=0.42r = 0.42r=0.42$) and outflow ($r=0.38r = 0.38r=0.38$) correlations explain the phenomena where greater volumes of traffic increases the chances of irregularities (Figure 6). This analysis improves the assistant's capability to focus onto features that sufficiently affect model performance for effective data preprocessing and enhanced model interpretability.

Overall, these findings confirm accuracy the intelligent assistant's capability in configuring data for urban transportation systems. Tackling some of the most critical issues like scalability, feature or anomaly detection, the assistant augments the reliability and efficiency of urban transportation analytics considerably. Further research will investigate the deployment of the assistant in real-life settings and how that would adapt the assistant to different data environments and scales for large smart city deployments.

Table 3 - Comparative Analysis of Data Preparation Technologies

Technology	Advantages	Disadvantages
Apache Spark	High processing speed and scalability.	Complex setup and limited adaptability.
Dask	Strong integration with Python and flexible usage	Requires manual configuration for new data types
Supervised ML Models	Effective with historical datasets, reducing manual intervention	Requires extensive labeled data and may lack generalization
Unsupervised ML Models	Does not require labeled data and is efficient in pattern recognition.	Limited interpretability and less accuracy in data cleaning.
Rule-Based Systems	Structured and reliable.	Poor scalability and struggles with complex datasets.
Intelligent Data Assistants	Effective for real-time data handling and adaptive learning.	Primarily focused on prediction rather than full automation of data preparation.

Table 3 provides a detailed comparative overview of the different technologies involved in data processing for urban transportation systems, focusing on the trade-offs between scalability, adaptability, and automation. In particular, Apache Spark and Dask stand out due to their ability to provide high performance and scalability for distributed data processing, but both require significant resources for manual tuning and lack dynamic adaptation to real-time changes. Supervised machine learning models excel at anomaly detection and classification but require large amounts of labeled data, while unsupervised models offer flexibility but have lower interpretability. Traditional rule-based systems provide structured verification but are not suitable for evolving complex data sets due to their poor scalability. Intelligent data assistants using machine learning and real-time analytics represent the most promising approach due to increased automation and adaptability, although they currently focus more on predictive analytics than on comprehensive data preparation. The results suggest that an optimal solution should integrate elements from multiple approaches, combining the efficiency of distributed frameworks, the adaptability of machine learning, and the robustness of rule-based validation. Future research should focus on improving intelligent assistants by implementing real-time learning mechanisms and enhancing automation capabilities to ensure robust data processing in dynamic urban environments.

Conclusion

In conclusion, this study emphasizes that the developed concept of an intelligent assistant for automating data preparation processes in urban transport management systems represents a fundamental breakthrough. Machine learning-based systems demonstrate the ability to significantly reduce manual intervention, improve data quality, and optimize urban transport operations, which has been confirmed by previous studies.

Future work includes implementation of an intelligent assistant prototype for practical application and testing in real conditions, evaluation of the system's effectiveness in various urban transport contexts, research and implementation of real-time data processing functions, which will allow for prompt response to changes, development of advanced predictive analytics capabilities for forecasting traffic flows and optimizing routes, adaptation of the system to a variety of urban conditions and transport infrastructure, ensuring its versatility and scalability.

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REFERENCES

- [1] Li B., Guo T., Li R., Wang Y., Gandomi A. H., Chen F. Self-Adaptive Predictive Passenger Flow Modeling for Large-Scale Railway Systems // IEEE Internet of Things Journal. - 2023. - Vol. 10(16). - P. 14182-14196. DOI 10.1109/JIOT.2023.3270427.
- [2] Iyer L. S. AI enabled applications towards intelligent transportation // Transportation Engineering. - 2021. - Vol. 5. - P. 100083. DOI 10.1016/j.treng.2021.100083.
- [3] Mahrez Z., Sabir E., Badidi E., Saad W., Sadik M. Smart urban mobility: When mobility systems meet smart data // IEEE Transactions on Intelligent Transportation Systems. - 2021. - Vol. 23(7). - P. 6222-6239. DOI 10.1109/TITS.2021.3084907.
- [4] Mohandu A., Kubendiran M. Survey on big data techniques in intelligent transportation system (ITS) // Materials Today: Proceedings. - 2021. - Vol. 47. - P. 8-17. DOI 10.1016/j.matpr.2021.03.479.
- [5] Paiva S., Ahad M. A., Tripathi G., Feroz N., Casalino G. Enabling technologies for urban smart mobility: Recent trends, opportunities and challenges // Sensors. - 2021. - Vol. 21(6). - P. 2143. DOI 10.3390/s21062143.
- [6] Oladimeji D., Gupta K., Kose N. A., Gundogan K., Ge L., Liang F. Smart transportation: an overview of technologies and applications // Sensors. - 2023. - Vol. 23(8). - P. 3880. DOI 10.3390/s23083880.
- [7] Ouallane A. A., Bakali A., Bahnasse A., Broumi S., Talea M. Fusion of engineering insights and emerging trends: Intelligent urban traffic management system // Information Fusion. - 2022. - Vol. 88. - P. 218-248. DOI 10.1016/j.inffus.2022.07.020.
- [8] Alahi M. E. E., Sukkuea A., Tina F. W., Nag A., Kurdthongmee W., Suwannarat K., Mukhopadhyay S. C. Integration of IoT-enabled technologies and artificial intelligence (AI) for smart city scenario: recent advancements and future trends // Sensors. - 2023. - Vol. 23(11). - P. 5206. DOI 10.3390/s23115206.
- [9] Yu G., Wang Y., Hu M., Shi L., Mao Z., Sugumaran V. RIOMS: An intelligent system for operation and maintenance of urban roads using spatio-temporal data in smart cities // Future Generation Computer Systems. - 2021. - Vol. 115. - P. 583-609. DOI 10.1016/j.future.2020.09.010.

[10] Lee J., Yoon Y. Indicators development to support intelligent road infrastructure in urban cities // Transport Policy. - 2021. - Vol. 114. - P. 252-265. DOI 10.1016/j.tranpol.2021.10.009.

[11] Autili M., Chen L., Englund C., Pompilio C., Tivoli M. Cooperative intelligent transport systems: Choreography-based urban traffic coordination // IEEE Transactions on Intelligent Transportation Systems. - 2021. - Vol. 22(4). - P. 2088-2099. DOI 10.1109/TITS.2021.3059394.

[12] Sundaresan S., Kumar K. S., Nishanth R., Robinson Y. H., Kumar A. J. Artificial intelligence and machine learning approaches for smart transportation in smart cities using blockchain architecture // Blockchain for Smart Cities. - Elsevier, 2021. - P. 35-56. DOI 10.1016/B978-0-12-824446-3.00009-0.

[13] Li B., Guo T., Li R., Wang Y., Gandomi A. H., Chen F. Self-Adaptive Predictive Passenger Flow Modeling for Large-Scale Railway Systems // IEEE Internet of Things Journal. - 2023. - Vol. 10(16). - P. 14182-14196. DOI 10.1109/IJOT.2023.3270427.

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ҚАЛАЛЫҚ КӨЛІК ЖҮЙЕЛЕРІНДЕ ДЕРЕКТЕРДІ ӨНДЕУДІ АВТОМАТТАНДЫРАТЫН ИНТЕЛЛЕКТУАЛДЫ АССИСТЕНТ

Аңдатпа. Қоғамдық көлік кестесін, қозғалыс сенсорларын және GPS құрылғыларын жинау және өңдеу қазіргі заманғы қалалық көлікті басқару жүйелері арқылы деректердің үлкен көлемін жасауға әкеледі. Кез келген аналитикалық немесе шешімге негізделген деректерді жасау үшін бұл деректерді жан-жақты тазалау, мүмкіндіктерді шығару және қалыпқа келтіру қажет. Бұл дәстүрлі әдістер жұмыс істеуі мүмкін болғанымен, олар көбінесе қымбат, дәлсіздіктерге ұшырайды және өте жалықтырады - әсіресе қалалық көлік жүйесінің үнемі өзгеретін динамикасымен айналысқанда. Нақты уақыттағы, үлкен көлемді тасымалдау деректері студенттерді тиімді және дәл ету үшін қосымша автоматтандыруды қажет етеді. Бұл құжат аясында машиналық оқыту алгоритмімен жасалған интеллектуалды көмекші қалалық көлік жүйелеріне деректерді дайындау үшін қажетті барлық жұмыстарды орындайды. Басқа интуитивті шешімдердің көпшілігі сияқты арнайы алдын ала өңдеу әрекеттеріне назар аударудың орнына, жүйе интеллектуалды автоматтандырылған мүмкіндікті шығаруды, деректерді онтайландыруды және аномалияны анықтауды қамтитын икемді және смарт құрылымды пайдаланады. Басқа он бір құрылым модульдікке кепілдік береді, бұл қол күшін азайтуға және есептеу тиімділігіне мүмкіндік береді, ақылды қала қолданбаларын мүмкін етеді. Қалаларға MDT өңдеуді автоматтандыруға, деректерді өңдеуді бақылауға кететін шығындарды азайтуға, тиімдірек басқаруға және азайған қателермен мұқият жұмыс істеуге көмектесетін жер үсті қоғамдық көлікті тиімдірек басқаруға және бағыттауға жаңа көзқарас ұсынылады.

Түйінді сөздер. Машиналық оқыту, интеллект, деректер, есептеулер, алгоритмдер, жүйе, автоматтандыру.

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ИНТЕЛЛЕКТУАЛЬНЫЙ АССИСТЕНТ ДЛЯ АВТОМАТИЗАЦИИ ПОДГОТОВКИ ДАННЫХ В СИСТЕМАХ УПРАВЛЕНИЯ ГОРОДСКИМ ТРАНСПОРТОМ

Аннотация. Сбор и обработка расписания общественного транспорта, датчиков дорожного движения и устройств GPS приведут к большому объему данных, генерируемых современными системами управления городским транспортом. Эти данные необходимо тщательно очищать, включать извлечение признаков и нормализацию для создания любых аналитических или основанных на решениях данных. Хотя эти традиционные методы могут работать, они часто являются дорогостоящими, подвержены неточностям и очень утомительными, особенно при работе с постоянно меняющейся динамикой городской транспортной системы. Крупномасштабные транспортные данные в реальном времени требуют дополнительной автоматизации, чтобы сделать студентов эффективными и точными. В этом документе интеллектуальный помощник, работающий с алгоритмом машинного обучения, выполнит всю необходимую работу по подготовке данных для городских транспортных систем. Вместо того чтобы фокусироваться на конкретных действиях предварительной обработки, как это делают большинство других интуитивных решений, система использует гибкую и интеллектуальную структуру, которая включает интеллектуальное автоматизированное извлечение признаков, оптимизацию данных и обнаружение аномалий. Одиннадцать других структур гарантируют модульность, которая позволяет сократить ручные усилия и повысить вычислительную эффективность, что делает возможными приложения для умного города. Предлагается новый подход к более эффективному контролю и управлению наземным общественным транспортом, который помогает городам автоматизировать обработку MDT, снижая затраты на обработку данных мониторинга, обеспечивая более эффективный контроль и аккуратную работу с меньшим количеством ошибок.

Ключевые слова. Машинное обучение, интеллект, данные, вычисления, алгоритм, система, автоматизация.

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